

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	5,470,000	5,209,524	(5,000,000)	(4,761,905)
2	5.113%	5.226%	5,470,000	4,950,807	(5,225,748)	(4,729,739)
3	5.312%	5.712%	5,470,000	4,683,288	(5,712,202)	(4,890,656)
4	5.494%	6.041%	0	0	0	0
				14,843,619		(14,382,300)
Fixed rate			5.470%	NPV	461,319	
Notional			100,000,000			

Figure 8.3: Computing the Roll-Down on the Swap

horizon date is exactly as it is today. For example, consider our stylized example and suppose we enter into some swap and look at a horizon date of one year. In order to examine the roll-down of this trade, the assumption we will make is that after entering into this trade, when we come back in one year, the swap curve is exactly as it was before (i.e., the 1-year swap rate is 5%, the 2-year swap rate is 5.11%, etc.). We do not literally expect this to happen, of course. Roll-down is an artificial construct that we use to analyze trades.

Once again, the roll-down of a swap is therefore the value of the swap as of the stated investment horizon assuming that the swap curve has not changed. It is the value of the swap after some period of time, recognizing that it has now aged and “rolled down” the curve.⁵ Consider again our 4-year swap in which we receive fixed. As of the investment horizon of one year, the original 4-year swap that we entered is now a 3-year swap with a coupon of 5.47%. This swap has aged one year and has now “rolled down” the curve; it was a 4-year swap with a coupon of 5.47%, and it is now a 3-year swap with a coupon of 5.47%. When we first entered into the swap, it had no value. Under the assumption that the curve has not changed, we are now in a 3-year swap with a coupon of 5.47% when the 3-year swap rate is 5.30%. The value of this swap, representing the roll-down on the trade, is calculated in Figure 8.3. As shown in this figure, the roll-down of the trade is \$461,319 on a notional of \$100mm, or approximately 46.13 bps. Thus, the total carry and roll-down for this trade is 47 bps + 46.13 bps $\approx$ 93.1 bps for a one year horizon.

⁵The swap curve is usually upward sloping. Thus as a swap ages, we can think of it like rolling down a hill, from which the phrase “roll-down” comes.

Time	zc	Forward	Period	bps Running	Cash Flow	PV Cash Flow
1	5.000%	5.000%	0x1			
2	5.113%	5.226%	1x2	17.4	174,262	157,721
3	5.312%	5.712%	2x3	17.4	174,262	149,199
4	5.494%	6.041%	3x4	17.4	174,262	140,699
					PV bps running	447,619
					PV carry	447,619
Notional		100,000,000			Net	0

Time	zc	Forward	Period	bps Running	Cash Flow	PV Cash Flow
1	5.000%	5.000%	0x1	17.0	170,000	161,905
2	5.113%	5.226%	1x2	17.0	170,000	153,864
3	5.312%	5.712%	2x3	17.0	170,000	145,550
4	5.494%	6.041%	3x4			
					PV bps running	461,319
					PV roll-down	461,319
Notional		100,000,000			Net	0

Figure 8.4: Converting the Carry and Roll-Down to Basis Points Running Over the Lives of the Respective Swaps

This quantity 93.1 bps represents the carry and roll-down as of the one year investment horizon. Many in the industry find it more intuitive to think of carry and roll-down instead in terms of bps running over the life of the underlying swaps.⁶ As shown in Figure 8.4, 44.76 upfront bps of carry is

⁶We need to be a little careful here because the two upfront numbers for the carry and roll-down are actually converted into bps running over the life of the same swap. By virtue of Equation 8.2, the carry is expressed as an annuity over the life of the residual swap. The roll-down, a measure of value of the aged swap, is expressed as an annuity over the life of the aged swap as of the horizon date. In our example of a 4-year swap and a one year horizon, this means that the upfront carry will be translated into bps running over the residual swap, which is a 1x4. And the upfront roll-down is translated into bps running over a 3-year spot starting swap as of the horizon date.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.344%	5.344%	5,470,000	5,192,512	(5,344,000)	(5,072,904)
2	5.457%	5.570%	5,470,000	4,918,543	(5,570,127)	(5,008,576)
3	5.657%	6.058%	5,470,000	4,637,587	(6,058,246)	(5,136,314)
4	5.840%	6.389%	0	0	0	0
				14,748,642		(15,217,794)
		Fixed rate	5.470%	NPV	(469,152)	
		Notional	100,000,000			

Figure 8.5: Mark-to-Market on Remaining \$100mm 3-Year Swap with a Coupon of 5.47% Assuming All Swap Rates Are 34.4 bps Higher in a Year Than They Are in Today's Swap Curve Given in Table 8.1

equivalent to 17.4 bps running over the 1x4 swap, and 46.13 upfront bps of roll-down is equivalent to 17.0 bps running over the 0x3 swap in a year.⁷

So what does this tell us? Carry and roll-down tells us how much we will make on the trade if rates are unchanged as of the horizon. Carry and roll-down expressed as bps running also tells us (approximately) how much we can be wrong and still break even on the trade. When we started this example and received fixed on a 4-year swap as an outright, macro trade, we were presumably expressing the view that the market would rally. But suppose we are wrong. Suppose that we receive fixed in a 4-year swap, and a year from now instead of the same swap curve prevailing, rates are all 34.4 bps higher than they are now. Then we should just about break even on our trade. If we put on this trade and in one year swap rates are 34.4 bps higher than they are now, the mark-to-market loss on the residual trade will be approximately offset by the carry earned in the first

⁷Actually, it is not necessary to go through the calculations in Figure 8.4 in order to translate the carry and roll-down into bps running. Let $s_{0,T}$ represent the par swap rate in a T -year swap. Consider a horizon t , where $t < T$. Denote by $s_{t,T}$ the par swap rate in a $T - t$ -year swap, t years forward, and denote by $s_{0,T-t}$ the par swap rate in a $T - t$ -year swap. Then the bps running for the carry is $s_{t,T} - s_{0,T}$, and the bps running for the roll-down is $s_{0,T} - s_{0,T-t}$. Let's consider now our stylized example with a 4-year swap and a one year investment horizon. Recall first that we showed in Chapter 3 that in this numerical example the 1x4 swap rate is 5.644%. Thus, the bps running for the carry is $5.644\% - 5.47\% = 17.4$ bps, and the bps running for the roll-down is $5.47\% - 5.30\% = 17$ bps, just as we found in Figure 8.4.

year.⁸ This phenomenon is demonstrated in Figure 8.5: assuming swap rates are 34.4 bps higher in a year than they are now, the mark-to-market on the remaining 3-year swap with a coupon of 5.47% is $-\$469,152$, which just about offsets the positive carry of $\$470,000$.

High carry and roll-down is a desirable attribute of a trade because it provides a buffer if the trade happens to go against us, not to mention a pure benefit if the market does nothing. If, for example, we receive fixed in a 4-year swap in hopes that the market will rally, and the market does nothing at all over the course of a year, we will be up by an amount equal to the nominal carry and roll-down. As was shown in Figure 8.3, the mark-to-market on the remaining trade is $\$461,319$, assuming rates are unchanged in this year. Adding this to the nominal carry earned in the first year gives a total of $\$461,319 + \$470,000 = \$931,319$.

This is not to suggest that investors necessarily put on trades solely for their carry and roll-down characteristics. But if an investor has a view on the market and finds a trade to express that view, high carry and roll-down will only serve to make the trade that much more attractive to him. Moreover, if an investor has a trade idea (e.g., receive fixed in swaps) in mind and is determining the best area to deploy it (e.g., determining what maturity swap to receive on), a comparison of carry and roll-down may be a deciding factor.

Figure 8.6 shows the carry and roll-down associated with the swap curve in Table 8.1 for a one year investment horizon. If an investor were thinking of receiving fixed somewhere in the 2- to 5-year sector but was indifferent as to maturity, he might use the results of Figure 8.6 and decide to receive fixed in 3-year swaps, since this maturity has the highest carry and roll-down. Alternatively, an investor who wanted to pay fixed (and will therefore be experiencing negative carry and roll-down) but was indifferent as to the exact maturity might decide to pay fixed in 2-year swaps, since this maturity has the lowest carry and roll-down.

8.2 Curve Trades

Sometimes people are interested in putting on trades that express a view on the difference between two different points on the yield curve. Such trades are generically referred to as *curve trades*. For example, a *2s10s swap curve*

⁸As mentioned previously, the bps running computations for carry and roll-down are computed over 3-year swaps, as indicated in Figure 8.4 by the two different boxed areas. The carry number is expressed as an annuity over a 3-year swap, one year forward. The roll-down number is expressed as an annuity over a spot 3-year swap, assuming rates are unchanged in a year. Thus, adding these two annuities is a bit of an apples and oranges exercise. Therefore the total annuity value will only serve as an approximate measure of how much of a sell-off we can endure and still break even.

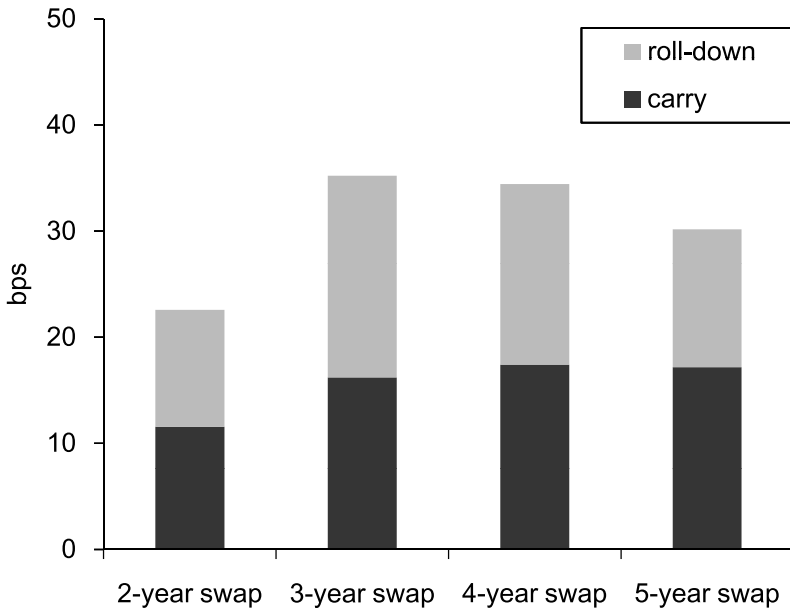


Figure 8.6: Comparison of Carry and Roll-Down (in bps Running) on Swaps of Different Maturities for a 1-Year Horizon, from the Perspective of the Fixed Rate Receiver, Based on the Swap Curve Given in Table 8.1

steepener (pronounced “twos tens”) is a trade in which someone is taking a view that the difference between the 10-year swap rate and the 2-year swap rate will increase. A *2s10s swap curve flattener* is a trade in which someone is taking the view that the difference between the 10-year swap rate and the 2-year swap rate will decrease.⁹

An investor will put on a curve trade by entering into two different swaps. If the investor receives fixed on the shorter-tenor swap and pays fixed on the longer-tenor swap, such a trade is a curve steepener, as the trade will increase in value if the difference between the longer-tenor swap rate and the shorter-tenor swap rate increases. Conversely, if the investor pays on the shorter-tenor swap and receives on the longer-tenor swap, such a trade is a curve flattener, as the trade will increase in value if the difference between the longer-tenor swap rate and the shorter-tenor swap rate decreases.

We will restrict ourselves to consider curve trades that are *duration-neutral*. This means that the PV01s on both swaps are chosen to be the same

⁹It is somewhat customary in the swaps market to identify a curve trade by specifying the shorter-tenor swap first, followed by the longer-tenor swap.

when the trade is initiated. This will require that different notional amounts are used on the two swaps. By choosing notionals such that the two swaps have the same PV01s, a change in either swap rate by a bp has the same dollar value for the investor who has put on the trade.¹⁰ In particular, this means that the investor is completely indifferent as to whether or not swap rates in general go up or go down; all that matters from the point of view of the investor putting on a duration-neutral curve trade is the difference in the two swap rates.¹¹

Why enter into curve trades? When an investor enters into a curve trade, he is expressing a view on the difference between two different points of the curve. Such a view may be driven by a combination of fundamental factors, macroeconomic factors, or technical factors. For example, suppose an investor believes that the Fed is going to have to initiate an unexpected and prolonged bout of lowering the Fed Funds rate in order to stimulate the economy before it enters into recession. At the same time, such easing from the Fed may raise concerns about the risk of inflation in the minds of investors. The investor may believe that the front end of the curve will likely rally as unanticipated rate cuts become known to the market and that the back end of the curve will likely sell off as the market comes to grips with the longer-term risk of inflation. The investor may want to express this view by entering into a 2s10s steepener.

Example

In order to delve into curve trades a bit more, let's turn our attention now to our stylized example and consider again the swap curve in Table 8.1. The current differential between the 5-year swap rate and the 2-year swap rate is $5.60\% - 5.11\% = 49$ bps. That is, 2s5s (in swaps) is 49 bps.¹² Suppose

¹⁰With the passage of time and changes in swap rates, a duration-neutral curve trade often does not remain duration-neutral. Many investors will readjust the notional on one or both legs of the trade periodically in order to maintain a duration-neutral trade.

¹¹Let's consider a particular example to elucidate this point. Suppose an investor enters into a 2s10s steepener and at the time of entry the 2-year swap rate is 5% and the 10-year swap rate is 6%. Consider the following three rate scenarios that may prevail shortly after the trade is put on: 1) the 2-year swap rate is 4.85%, and the 10-year swap rate is 5.95%; 2) the 2-year swap rate is 5.05%, and the 10-year swap rate is 6.15%; and 3) the 2-year swap rate is 4.95%, and the 10-year swap rate is 6.05%. In a duration-neutral curve trade, the investor is indifferent to any of the three scenarios. In each case the 2s10s curve has steepened 10 bps and is now at 110 bps. The first scenario happened to occur as the market rallied; the second, as the market sold off; and the third, where the front end rallied and the back end sold off.

¹²Some investors prefer to look at curve trades involving other yield curves, such as the Treasury curve. If it is clear from context which curve we are talking about, then we might just identify which points on the curve we are discussing by saying something

we think that 2s5s will steepen from its current level and want to put on a trade that risks \$100,000 per bp.

We will want to enter into a 2-year spot starting swap and a 5-year spot starting swap. We will need to duration weight the trade, such that a bp movement in either the 2-year or the 5-year swap rate is worth \$100,000. Recall from Section 2.1.1 that there are a number of ways to consider the PV01 of a swap. We will choose to compute the PV01s using the “PV Coupon 01” approach.¹³ As shown in Figure 8.7, the notional weightings that give a PV01 of \$100,000 on each leg are (approximately) \$233.57mm of the 5-year swap and \$538.37mm of the 2-year swap.¹⁴ If we want to express the view that the curve is going to steepen, we should pay in 5s at a rate of 5.60% and receive in 2s at a rate of 5.11%.¹⁵

Now suppose that at the end of the same day the swap curve is as in Table 8.2. The 2s5s curve is now $5.65\% - 5.18\% = 47$ bps and has thus flattened 2 bps on the day. Since we put this trade on for \$100,000 per bp, we would expect to be down \$200,000. In order to value the steepener at the end of the day based on these levels, we must value each leg separately, as shown in Figure 8.8. In the steepener, we paid in 5s at 5.60%, and 5-year swap rates ended the day at 5.65%. So we have made 5 bps (running) on this leg of the trade, and as shown in Figure 8.8 the value of the leg is \$499,389. We also received in 2s at 5.11%, and 2-year swap rates ended the day at 5.18%, so we lost 7 bps (running) on this leg. As shown in Figure 8.8,

like “2s5s.” If there is any ambiguity regarding which curve we are discussing, we might say something like “2s5s in swaps” or “2s5s in Treasuries.” There are even times when investors enter into curve trades in which one leg is put on in Treasuries and the other leg is in swaps. For example, suppose an investor wants to enter into a curve steepener, believing that there will be a flight-to-quality bid in the front end of the Treasury curve (meaning buying of short-dated Treasuries will cause short-end yields to come down). At the same time the investor believes that the flight-to-quality will cause swap spreads across the curve to increase. If the investor is thinking of putting on a 2s5s steepener, he may want to consider buying 2-year Treasury notes and paying fixed in a 5-year swap.

¹³This is a little sloppy. It would be more correct to base notionals on PV01s based on a change in the swap curve; however, the two numbers are very close and as such many investors choose to use the “PV Coupon 01” numbers for ease.

¹⁴In Figure 8.7 we have solved for the notional on the 2-year swap such that when we move the fixed coupon off market by 1 bp (from 5.11% to 5.12%), the value of the swap is the desired \$100,000. We do a similar thing for the 5-year leg.

¹⁵An easy way to figure out which way to go around in order to put on a steepener is to hold one leg constant, say, the 2-year, and then ask which way we have to go on the 5-year in order to make money if the curve steepens. If we are holding the 2-year constant, then for the curve to steepen means that the 5-year swap rate goes up. In order to benefit if this happens, we will want to pay in 5s.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	13,103,532	12,479,554	11,678,727	11,122,597
2	5.113%	5.226%	13,103,532	11,859,791	12,206,018	11,047,466
3	5.312%	5.712%	13,103,532	11,218,943	13,342,251	11,423,328
4	5.494%	6.041%	13,103,532	10,579,780	14,111,089	11,393,281
5	5.635%	6.202%	13,103,532	9,961,932	14,486,496	11,013,328
				56,100,000		56,000,000
Fixed rate			5.610%	NPV	100,000	
Notional			233,574,545			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	27,564,457	26,251,863	26,918,415	25,636,585
2	5.113%	5.226%	27,564,457	24,948,137	28,133,772	25,463,415
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
				51,200,000		51,100,000
Fixed rate			5.120%	NPV	100,000	
Notional			538,368,293			

Figure 8.7: Computing the Notionals on Each Leg of the 2s5s Steepener for a Duration-Neutral Trade with a PV01 of \$100,000

the value of this leg is against us \$699,307. Thus, our P&L on the trade is $\$499,389 - \$699,307 = -\$199,918$.¹⁶ □

¹⁶Note that our loss is not exactly \$200,000, which is the number we expected, but very close to that amount. The reason has to do with the fact that we chose the “PV Coupon 01” methodology for choosing notionals, not to mention the fact that the (albeit quite small) shift in rates has caused the PV01 of each leg of the trade to change slightly. As indicated by our P&L number, the error is quite small.

Maturity	Swap Rate
1 Year	5.07%
2 Year	5.18%
3 Year	5.33%
4 Year	5.50%
5 Year	5.65%

Table 8.2: The Swap Curve at the End of the Same Day

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.070%	5.070%	(13,080,175)	(12,449,010)	11,842,229	11,270,800
2	5.183%	5.296%	(13,080,175)	(11,822,890)	12,369,700	11,180,707
3	5.340%	5.655%	(13,080,175)	(11,190,052)	13,209,479	11,300,672
4	5.522%	6.071%	(13,080,175)	(10,549,542)	14,181,361	11,437,681
5	5.687%	6.346%	(13,080,175)	(9,920,043)	14,822,022	11,241,065
				(55,931,537)		56,430,926
		Fixed rate	5.600%	NPV	499,389	
		Notional	233,574,545			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.070%	5.070%	27,510,620	26,183,135	(27,295,272)	(25,978,179)
2	5.183%	5.296%	27,510,620	24,866,261	(28,511,045)	(25,770,524)
3	5.340%	5.655%	0	0	0	0
4	5.522%	6.071%	0	0	0	0
5	5.687%	6.346%	0	0	0	0
				51,049,396		(51,748,703)
		Fixed rate	5.110%	NPV	(699,307)	
		Notional	538,368,293			

Figure 8.8: Computing the Value of the 2s5s Steepener at the End of the First Day Based on the Curve Shown in Table 8.2

Maturity	Swap Rate
1 Year	5.10%
2 Year	5.23%
3 Year	5.40%
4 Year	5.62%
5 Year	5.75%

Table 8.3: The Swap Curve After One Year

8.2.1 Yield Curve Trades for Longer Holding Periods

In the previous section, we calculated the P&L of a curve trade on the same day that we had put it on. Oftentimes an investor who puts on a curve trade will anticipate keeping the trade on for some holding period as he waits for his view to come to fruition. For example, in the 2s5s steepener that we just examined, the P&L was negative at the end of the first day. An investor would likely continue to hold the trade, provided that he still has conviction in his view of the curve steepening and provided that he did not hit his *stop-loss* on the trade.¹⁷ In this case, the carry and roll-down characteristics of the trade may be something that the investor wants to consider prior to putting it on.

Let's go back to our original swap curve as shown in Table 8.1 and assume that we put on the 2s5s steepener for \$100,000 per bp by paying on \$233.57mm of the 5-year swap and receiving on \$538.37mm of the 2-year swap. We are in the trade at $5.60\% - 5.11\% = 49$ bps and are hoping the 2s5s curve steepens. Suppose now that instead of taking the position off at the end of the first day, we hold the position for one year. Suppose further that after one year the swap curve is as shown in Table 8.3. Notice that after one year the 2s5s curve is $5.75\% - 5.23\% = 52$ bps. But we should be a little careful here. Since one year has gone by, the 2-year swap that we traded is now a 1-year swap, and the 5-year swap that we traded is now a 4-year swap. Thus, the 2s5s steepener that we put on at 49 bps is now a 1s4s steepener at 49 bps. And, from Table 8.3, the 1s4s curve is currently $5.62\% - 5.10\% = 52$ bps. The 1s4s curve is indeed the relevant benchmark to consider after one year, because as of this time each of the two swaps has bled its first-year coupons, and our trade is now a 1s4s steepener.

Suppose that we want to unwind our trade now based on these levels. In order to compute our P&L on the trade, we need to compute both the carry

¹⁷A stop-loss is a predetermined mark-to-market loss that a trader establishes on a trade that, if reached, will prompt the trader to get out of the trade so as to limit any further downside.